PURE PROJECTIVE MODULES AND UNIVERSALLY EXACT RESOLUTIONS BY DIRECT SUMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The criterion for universal exactness of a sequence of modules, and the characterisation of pure projective modules. Algebra, Theorem algebra-theorem-universally-exact-criteria, invoked at proof lines 2694-2695 and 2706-2708. Makes finitely presented modules pure projective, shows the constructed sequence is universally exact, and turns splitting into pure projectivity.

Source reference. Original theorem 058K: sources/stacks-chow/tags.txt, source lines 6783–6783. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring.*

(1) *A module is pure projective if and only if it is a direct summand of a direct sum of finitely presented A-modules.*
(2) *For any module M there exists a universally exact sequence $0 \to N \to P \to M \to 0$ with P pure projective.*

Proof. First note that a finitely presented A-module is pure projective by Algebra, Theorem 058K. Hence a direct summand of a direct sum of finitely presented A-modules is indeed pure projective. Let M be any A-module. Write $M = \operatorname{colim}_{i \in I} P_i$ as a filtered colimit of finitely presented A-modules. Consider the sequence

$$0 \to N \to \bigoplus P_i \to M \to 0.$$

For any finitely presented A-module P the map $\mathrm{Hom}_A(P, \bigoplus P_i) \to \mathrm{Hom}_A(P, M)$ is surjective, as any map $P \to M$ factors through some P_i. Hence by Algebra, Theorem 058K this sequence is universally exact. This proves (2). If now M is pure projective, then the sequence is split and we see that M is a direct summand of $\bigoplus P_i$. □

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